B224-101-1112

101-E-102A/B/C/D
CRUDE/CRUDE COLUMN OVERHEAD EXCHANGER

TYPE-C, NOTE-1
S/S DT - 175° C
DP - 36.7 kg/cm²g
T/S DT - 161.5° C
DP - 28.2 kg/cm²g
DESIGN DUTY - 26.74 MMkcal/hr

FLARE 101-1177 — 4" -FL-101-1204-A11A — 6"-FL-101-1205-A11A

FROM 101-E-130A/B WASH WATER 101-1148 — 6"-WP-101-4811-B19A-IH

FROM 101-C-101 CR COL OVHD VAP 101-1127 — 48"-P-101-2701-A19A-IH

24"-P-101-1209-B1A-IT TO 101-E-103A/B & 105A/B CRUDE 101-1113

FROM 101-E-101A/B CRUDE 101-1111 — 24"-P-101-1113-B1A-IT (NOTE-6)

36"-P-101-1210-A19A-IH (NOTE-3,5) — 36"x40" — 40"x36" — 48"x40" — 48"x40" — NOTE-3,5 — 36"-P-101-1211-A19A-IH

MAKE CONNECTION AT MID POINT OF HEADER FROM TOP

NO POCKETS

14"x20" — 20"x24" — NOTE-6 — 24"x16" — 16"x12"

26"-P-101-1212-A19A-IH — 12"-P-101-1205-B1A-IT — 12"-P-101-1206-B1A-IT — 26"-P-101-1213-A19A-IH — 2"-WP-101-1202-B19A-IH — 3"-FL-101-1202-A11A — 3"-FL-101-1203-A11A — 26"-P-101-1214-A19A-IH — 2"-WP-101-1203-B19A-IH — 12"-P-101-1207-B1A-IT — 12"-P-101-1208-B1A-IT — 26"-P-101-1215-A19A-IH — 2"-WP-101-1204-B19A-IH — 3"-FL-101-1204-A11A

SLOPE 1:50

FE-1201 FE-1202 FE-1203 FE-1204 — FACING

2"-P-101-1220-B1A-IT — 2"-P-101-1221-B1A-IT — 2"-P-101-1222-B1A-IT — 2"-P-101-1223-B1A-IT

2"-SV-101-1201-B1A-IT — 2"-SV-101-1202-B1A-IT — 2"-SV-101-1203-B1A-IT — 2"-SV-101-1204-B1A-IT — 1.5E2 86.7 — PSV 1201 1202 1203 1204 — 300# — 2"x1-1/2" — 3/4" — LG FB

FOD101111201 — FOD101111202 — FOD101111203 — FOD101111204 — FOD101111205 — FOD101111206 — FOD101111207 — FOD101111208 — FOD101111209 — FOD101111210 — FOD101111211 — FOD101111212

26"x24" — T1 — T2 — S1 — S2 — 300# — A — B — C — D — 101-E-102A/B/C/D

MIN — NOTE-8 (TYP.) — NOTE-4 — A19A B19A — A11A A19A H — 3/4"

SP 110 — SP 111 — SP 112 — SP D10 — SP D12 — SP D14 — SP D16 — SP D18 — SP D36 — SP D08

CO — CR 1210 — CR 1209 — CR 1207 — CR 1208 — CR 1212 — CR 1215 — CR 1216

TW 1212 — PI 1205 — TE 1223 — TI 1223 — FF — TI 1223

TI 1201 — TE 1201 — FI 1201 A — FI 1201 B — FT 1201

TI 1202 — TE 1202 — FI 1202 A — FI 1202 B — FT 1202

TI 1203 — TE 1203 — FI 1203 A — FI 1203 B — FT 1203

TI 1204 — TE 1204 — FI 1204 A — FI 1204 B — FT 1204

TI 1205 — TE 1205 — FI 1205 — TW 1210 — PI 1201 — DR

TI 1206 — TE 1206 — FI 1206 — TI 1207 — TE 1207 — FI 1207 — TI 1208 — TE 1208 — FI 1208 — TI 1209 — TE 1209 — FI 1209 — FI 1212 — FI 1215

TWO PHASE FLOW — NOTE-5,10 — NO POCKETS

24"-P-101-1219-A19A-IS — TO 101-A-101 D — CR COL OVHD 101-1128

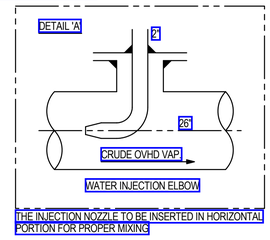
24"-P-101-1218-A19A-IS — TO 101-A-101 C — CR COL OVHD 101-1128

24"-P-101-1217-A19A-IS — TO 101-A-101 B — CR COL OVHD 101-1128

24"-P-101-1216-A19A-IS — TO 101-A-101 A — CR COL OVHD 101-1128

NOTES: -

1. FOR LEGEND SYMBOL AND GENERAL DETAILS, REFER P&ID NO. B068-79-41-00-1191/1192 (TPL DWG. NO. B224-CDU-TPL-000-1191/1192-SHT01/1192-SHT02)
   - FOR EXCHANGER DRAIN DETAILS REFER P&ID NO. B224-CDU-TPL-101-1246-SHT01 & B224-CDU-TPL-101-1246-SHT02
   - FOR AIR COOLER DRAIN CONNECTION DETAILS REFER P&ID NO. B224-CDU-TPL-101-1247
   - FOR PUMP DRAIN/VENT REFER P&ID NO.B224-CDU-TPL-101-1245-SHT01 & B224-CDU-TPL-101-1245-SHT02
   - FOR FILTER DRAIN VENT DETAILS REFER P&ID NO. B224-CDU-TPL-101-1248
   - FOR PUMP SEAL PLAN DETAIL REFER P&ID NO. B224-CDU-TPL-101-1184-SHT01 B224-CDU-TPL-101-1184-SHT02 & B224-CDU-TPL-101-1184-SHT03
   - FOR CONTROL VALVE DRAIN VENT DETAILS REFER P&ID NO. B224-CDU-TPL-101-1243-SHT01 & B224-CDU-TPL-101-1243-SHT02
   - FOR LEVEL INSTRUMENT DRAIN VENT DETAILS REFER P&ID NO. B224-CDU-TPL-101-1244-SHT01 & B224-CDU-TPL-101-1244-SHT02
   - ALL INSTRUMENT TAG NUMBERS TO HAVE THE UNIT DESIGNATION PREFIX 101
2. ALL DCS OPEN AND CLOSED LOOP INSTRUMENTS SHOWN IN P&ID WITH 'FF' ARE FOUNDATION FIELD BUS DEVICES. FOR INSTRUMENTS FOR WHICH FIELD BUS DEVICES ARE NOT SHOWN/AVAILABLE, SMART TRANSMITTERS/SMART POSITIONERS SHALL BE USED.
3. SYMMETRICAL PIPING.
4. WASH WATER TO BE INJECTED WITH REVERSE FLOW, CONE SPRAY INJECTION NOZZLE AS PER DETAIL SHOWN BELOW.



THE INJECTION NOZZLE TO BE INSERTED IN HORIZONTAL PORTION FOR PROPER MIXING.

5. MAXIMUM ALLOWABLE PRESSURE DROP IN THE LINES IN THE CRUDE COLUMN OVHD CIRCUIT (I.E. FROM 101-C-101 TO 101-V-101 VIA 101-E-102A/B/C/D AND 101-A-101) IS 0.1 KG/CM2. ALL LINE SIZES IN THE CIRCUIT ARE CONFIRMED BASED ON FINAL ISOMETRICS.
6. CRUDE INLET AND OUTLET LINES TO BE ORIENTED SUCH THAT THE INLET AND OUTLET ARE DIAGONALLY OPPOSITE TO EACH OTHER AS SHOWN ON P&ID.
7. CORROSION MONITORING SYSTEM (CMS) SHALL BE DONE WITH CORROSION PROBE & CORROSION COUPON.
8. A MIN. OF 1 TO 1.5 M DISTANCE SHOULD BE KEPT BETWEEN CP AND CC. FOR MAINTENANCE PURPOSE.
9. CHEMICAL INJECTION SHALL BE DOSED THROUGH INJECTION TUBE WITH T-TYPE ACCESS FITTINGS.
10. LINE TO BE PROPERLY SUPPORTED.

RED = Equipment Info
BLUE = Process Tags

No	REVISIONS	DATE	DRN	BY	CKD	APPD
4	ISSUED FOR CONSTRUCTION	10.03.22	SG	MW	VD	PV
3	ISSUED FOR CONSTRUCTION	30.09.21	SG	MW	VD	PV
2	ISSUED FOR ENGG/ DESIGN WITH HAZOP RECOMMENDATION	02.09.20	RMB	KG	RS	SG
1	ISSUED FOR ENGINEERING/DESIGN	11.08.20	JG	KG	RS	SG
0	ISSUED FOR REVIEW	13.03.20	LG	KG	RS	SG

DETAILED ENGINEERING CONTRACTOR: FLUOR — FLUOR DANIEL INTERNATIONAL (MALAYSIA) SDN BHD

LSTK CONTRACTOR: TATA PROJECTS LIMITED, MUMBAI

No	DATE	REVISIONS	DRN	BY	CKD	APPD
B	04.12.18	REVISED AS MARKED	PP	MATA	GN	KM
A	24.11.18	ISSUED FOR PACKAGE	SKU	MATA	GN	KM

PMC: इंजिनियर्स इंडिया लिमिटेड (भारत सरकार का उपक्रम) — ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

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पाइपिंग एण्ड इंस्ट्रूमेन्टेशन डायाग्राम
PIPING AND INSTRUMENTATION DIAGRAM
CRUDE/VACUUM DISTILLATION UNIT
CRUDE PREHEAT TRAIN-1B

अनुमाप SCALE	कार्य संख्या JOB NO.	विभाग DEPT.	अनुभाग SECTN.	इकाई UNIT	आरेख संख्या DWG. No.	संशो REV.
	B068	79	41	101	1112	1

TPL DRAWING NO. / टी पी एल आरेख संख्या: B224-CDU-TPL-101-1112 — संशो REV. 4

DOCUMENT REVIEW DETAILS

V	Vendor Comments.
2	Comply with manufacture / fabrication as per the reviewed document. Revised document required.
3	Document does not conform to basic requirements as marked. Resubmit for review.
R	Returned for Records.
V	Void.

Doc. B224-CDU-TPL-101-1112 Rev. 4
Signature with Date: 24-AUG-2022
Name of Reviewer: UMER AFZAL (C889)
Page No.: 1 of 2

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